Global Government Policies and Climate Change: A Literature-Based Review of Policy Approaches and Effectiveness

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Abstract

This literature review looks at the findings from six scholarly sources examining various facets of global government responses to climate change. The studies cover national policy actions, media influence, public perception, economic modeling, fossil fuel dependency, and maladaptation. Overall, the literature reveals a complex interplay between socio-political dynamics and environmental strategy, underscoring the need for evidence-based, adaptable, and forward-thinking approaches. From Indonesia's national policy alignment with research to U.S. legislative responsiveness to media coverage, the research illustrates both promising developments and persistent barriers to effective climate governance. The review highlights how uncertainty, social preferences, and misaligned incentives complicate decision-making, emphasizing the necessity of interdisciplinary solutions.

Introduction

Climate change represents one of the most pressing global challenges of our time, and governments worldwide have approached this threat through varying policies and frameworks. A common thread across the literature is the tension between scientific understanding, public opinion, and political will. Dolšak and Houston (2014) explored how U.S. media coverage influences legislative action, while Djalante, Jupesta, and Aldrian (2021) analyzed Indonesia's efforts to synchronize national policies with scientific research. Höök and Tang (2013) examined the connection between fossil fuel depletion and climate policy development, adding a geological dimension to climate strategy. Pindyck (2013) critiqued the economic models that underlie major policy decisions, revealing how uncertainty hinders effective planning. Kennard (2021) brought attention to psychological and social influences, showing how concern for others affects climate engagement. Finally, Sarkar, Bandyopadhyay, and Singh (2024) discussed the dangers of maladaptation and the systemic risks associated with policy missteps. Together, these works offer a multidimensional view of how governments conceptualize and combat climate change.

Discussion

Climate policy in democratic societies often hinges on public perception and media framing. Rather than operating solely on scientific urgency, legislators respond to what the media highlights and how constituents react. Dolšak and Houston (2014) found that U.S. newspaper coverage positively correlates with the introduction of climate-related legislation. Their analysis suggested that higher media visibility of climate issues results in more congressional activity, especially when coverage emphasizes risks and presents consensus on anthropogenic causes. However, the study also pointed out that not all media is equally influential, regional differences in reporting and political orientation of newspapers can skew public perception and legislative outcomes.

In contrast, Djalante et al. (2021) examined Indonesia's systematic attempt to integrate climate science into national policy. Indonesia's National Action Plan for Climate Change (RAN-GRK) is an example of how governments can harmonize scientific research and legislative action. The authors praised the participatory approach involving local governments and stakeholders, which helped identify practical mitigation strategies. However, they also cautioned that funding limitations and administrative fragmentation posed ongoing challenges. Their study highlights the importance of institutional coordination and capacity-building in implementing climate strategies in developing nations.

Fossil fuel availability is a foundational issue in climate policy, yet it is often treated as separate from climate debates. Höök and Tang (2013) argued that depletion of fossil fuels, especially oil, could unintentionally aid climate goals by forcing reductions in consumption. However, they warned that without proactive planning, this depletion could result in energy insecurity and reliance on alternative high-emission fuels like coal. The paper called for policies that simultaneously address resource limits and environmental protection, a dual challenge that many governments have yet to confront cohesively.

When it comes to modeling climate policy, uncertainty is a major obstacle. Pindyck (2013) criticized the Integrated Assessment Models (IAMs) used to guide carbon pricing and emissions targets, arguing that they are based on arbitrary assumptions and offer limited practical insight. He noted that climate change involves deep uncertainty, especially regarding long-term temperature impacts and damage functions, which makes these models both overly confident and misleading. Pindyck advocated for simpler, more transparent approaches grounded in risk management rather than pseudo-scientific precision, highlighting a gap between academic modeling and policymaker needs.

Social values and individual psychology also shape how societies approach climate change. Kennard (2021) explored how "other-regarding preferences," or concern for the well-being of others, influence support for climate policy. Using survey data across multiple countries, the study found that people who prioritize the welfare of future generations or disadvantaged groups are more likely to support mitigation efforts. This suggests that policy communication strategies could benefit from emphasizing moral responsibility and collective benefits rather than just self-interest or economic costs.

Finally, the issue of maladaptation, when policies intended to help actually exacerbate vulnerability, is gaining attention. Sarkar, Bandyopadhyay, and Singh (2024) outlined how climate strategies, when based on limited data or short-term political goals, can worsen inequality and environmental risk. For example, building sea walls may protect wealthier urban zones while increasing flood risk in surrounding rural areas. Their research emphasized the need

for adaptive, inclusive, and system-aware policymaking that considers long-term feedback loops and unintended consequences.

Conclusion

The research shows that successful climate policy is not just about having the right data or models. It also depends on how well governments can connect science with public values, social priorities, and political realities. Some countries, like Indonesia, are making progress by aligning research with policy and involving local stakeholders. In contrast, other nations struggle due to media influence, public skepticism, or overly complex economic models that miss the bigger picture. These studies make it clear that climate change is not only an environmental or scientific problem. It is also a political, social, and moral challenge. Moving forward, governments will need to focus on collaboration, transparency, and long-term thinking to create fair and effective solutions that actually work for people and the planet.

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